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SENIOR DATA ENGINEER

PROFESSIONAL SUMMARY

- Around 6 years of experience in the IT industry and expertise in Big Data/ Hadoop Development framework and Analysis, Design, Development, Testing, Documentation, Deployment, and Integration using SQL, Python, Spark, and Big Data technologies.
- Experience in software analysis, design, development, testing, and implementation of Big Data, Hadoop, SQL, and No SQL technologies.
- A very good understanding of job workflow scheduling and monitoring tools like Oozie and Control-M.
- Experience in the Software Development Life Cycle (SDLC) phases which include Analysis, Design, Implementation, Testing, and Maintenance.
- Strong technical, administration, and mentoring knowledge in Linux and Big data/Hadoop technologies.
- Extensively used Azure Data Lake and worked with different file formats csv, Avro, parquet, orc, and delta.
- Documented business, process, and data flow documentation on multiple data systems.
- Designed Data models on multiple database systems, customer support, troubleshooting, and modifying databases to meet the customer’s requirements.
- Written Terraform scripts to automate AWS services which include CloudFront distribution, RDS, EC2, and S3 Buckets.
- Implemented Lambda architecture using Azure Data platform capabilities like Azure Data Lake, Azure Data Factory, HDInsight, Azure SQL Server, Azure ML, and Power BI.
- Expertise in Snowflake to create and Maintain Tables and views.
- Hands-on experience with multiple ML/AI Algorithms such as Neural networks, Linear and Logistic Regression, Support Vector Machine, Decision Tree, Naive Bayes, Apriori, K means clustering, KNN classifier using R and Python libraries (git, Beautiful Soup, NumPy, Matplotlib, Whoosh, PySpark) to perform predictive Analytics on various datasets.
- Good working experience in using Spark SQL to manipulate Data Frames in Python.
- Good knowledge of NoSQL databases including Cassandra and MongoDB.
- Expertise in Data warehouses like Azure Synapse and Snowflake.
- Experience in NoSQL Column-Oriented Databases like HBase and its Integration with Hadoop cluster.
- Knowledge of implementing Big Data in Amazon Elastic MapReduce (Amazon EMR) for processing, and managing Hadoop framework dynamically scalable Amazon EC2.
- Develop ETL mappings for various Sources (TXT, .CSV) and load the data from these sources into relational tables with Talend Enterprise Edition.
- Experience working with Azure BLOB and Data Lake storage and loading data into Azure SQL Synapse analytics (DW).
- Experience with Amazon Web In-depth knowledge of Hadoop Architecture and various components such as HDFS, Job Tracker, Task Tracker, Name Node, Data Node, and MapReduce concepts.
- Services, AWS command line interface, and AWS data pipeline.
- Experience in writing SQL, PL/SQL queries, and Stored Procedures for accessing and managing databases such as Oracle, ANSI SQL, PostgreSQL, MySQL, Amazon Aurora, Azure Synapse, and IBM DB2.
- Experience in Developing Spark applications using Spark-SQL in Databricks for data extraction, transformation, and aggregation from multiple file formats for analyzing & transforming the data to uncover insights into customer usage patterns.
- Developed Power BI reports & effective dashboards after gathering and translating end-user requirements.
- Used various sources to pull data into Power BI such as SQL Server, Excel, Oracle, SQL Azure, etc.
- Experience in Scripting languages like Bash, Unix Shell scripting and knowledge of Python

- Good working knowledge of analysis tools like Tableau for regression analysis, pie charts, and bar graphs.
- Skilled in provisioning, configuring, and maintaining Amazon EC2 instances for various data processing tasks and analytics workloads.
- Expertise in developing Spark code using Scala and Spark-SQL/Streaming for faster testing and processing of data.
- Experience with Apache Spark ecosystem using Spark-SQL, Data Frames, RDDs, and knowledge of Spark MLlib.
- Hands on experience with Big Data Ecosystems including Hadoop, MapReduce, Pig, Hive, Impala, Sqoop, Flume, NIFI, Oozie, MongoDB, Zookeeper, Kafka, Maven, Spark, Scala, HBase, Cassandra.
- Designed and implemented custom NiFi processors that reacted, and processed for the data pipeline.
- Experience in installation, configuration, and deployment of Big Data solutions.
- Good usage of Apache Hadoop along with the enterprise version of Cloudera and Hortonworks.
- Extensive experience in developing stored procedures, functions, Views and Triggers, and Complex queries using SQL Server, TSQL, and Oracle PL/SQL
- Experience with Data Cleansing, Data Profiling, and Data analysis. UNIX Shell Scripting, SQL, and PL/SQL coding.
- Database / ETL Performance Tuning: Broad Experience in Database Development including effective use of Database objects, SQL Trace, Explain Plan, Different types of Optimizers, Hints, Indexes, Table Partitions, Sub Partitions, Materialized Views, Global Temporary tables, Autonomous Transitions, Bulk Binds, Capabilities of using Oracle Built-in Functions. Performance Tuning of Informatica Mapping and Workflow.
- Superior communication skills, strong decision-making and organizational skills along with outstanding analytical and problem-solving skills to undertake challenging jobs. Able to work well independently and in a team by helping to troubleshoot technology and business-related problems.

TECHNICAL SKILLS

Big Data Ecosystem:	HDFS, Yarn, MapReduce, Spark, Kafka, Kafka Connect, Hive, Airflow, Stream Sets, Sqoop, HBase, Flume, Pig, Oozie
Hadoop Distributions:	Hadoop, Cloudera CDP, Hortonworks HDP
Cloud Environment:	Amazon AWS - EMR, EC2, EBS, RDS, S3, Athena, Lambda, SQS, DynamoDB, RedShift, Azure, GCP
Scripting Languages:	Python, Java, Scala, R, PowerShell Scripting, Pig Latin, HiveQL
NoSQL Database:	HBase, MongoDB
Database:	MySQL, Oracle, Teradata, MS SQL SERVER, PostgreSQL, DB2
ETL/BI:	Snowflake, Informatica, Talend, SSIS, SSRS, SSAS, ER Studio, Tableau, PowerBI
Operating Systems:	Linux (Ubuntu, Centos, RedHat), Windows (XP/7/8/10/11)
Others:	Docker, Spring Boot, JIRA

PROFESSIONAL EXPERIENCE

Client: SMBC
Sept 2023 – Present
Role: Sr. Data Engineer

Responsibilities:

- Responsible for gathering requirements, understanding business values, and providing analytical data solutions.
- Worked on the AWS Data pipeline to configure data loads from S3 into Redshift.
- Using AWS Redshift, I Extracted, transformed, and loaded data from various heterogeneous data sources and destinations Created Tables, Stored Procedures, and extracted data using T-SQL for business users whenever required.
- Designed and implemented data ingestion pipelines using AWS Glue, ensuring efficient and scalable data processing.
- Leveraged AWS data services such as Amazon RDS, Amazon Redshift, and Amazon Athena for designing, implementing, and managing databases.
- Worked with AWS Glue for ETL (Extract, Transform, Load) processes, ensuring efficient data movement and transformation.
- Developed and maintained PL/SQL stored procedures, functions, triggers, and packages to support data operations within AWS databases.
- Developed ETL (Extract, Transform, Load) workflows in AWS Glue to cleanse, transform, and load data from various sources into data lakes and data warehouses.
- Developed Spark applications using Pyspark and Spark-SQL for data extraction, transformation, and aggregation from multiple file formats.
- Utilized AWS Glue Dynamic Frames for schema inference and dynamic data transformations, improving data processing flexibility.
- Managed and optimized AWS Glue Data Catalog to maintain metadata and improve query performance in Athena.
- Created and managed AWS Glue Crawlers to automatically discover and catalog metadata from data sources in Amazon S3.
- Implemented data partitioning strategies in Amazon S3 to enhance query performance and reduce data processing costs.
- Developed custom PySpark scripts within AWS Glue Jobs for complex data transformations and data enrichment.
- Worked on data lineage tracking and data quality monitoring using AWS Glue and AWS CloudWatch.
- Implemented data encryption and security measures for sensitive data stored in Amazon S3.
- Automated data pipeline orchestration using AWS Step Functions and Lambda functions.
- Developed AWS Lambda functions for event-driven data processing and data pipeline triggers.
- Collaborated with DevOps teams to integrate AWS Glue into CI/CD pipelines for automated deployment and monitoring.
- Designed and maintained data lake architectures in Amazon S3, incorporating data partitioning and lifecycle policies.
- Authoring Python (PySpark) Scripts for custom UDFs for Row/ Column manipulations, merges, aggregations, stacking, data labeling, and all Cleaning and conforming tasks.
- Developed and optimized Spark jobs using PySpark to extract data from AWS S3, transform it using SQL and Spark functions, and write it to Redshift.
- Created and managed AWS Athena queries to enable self-service data exploration for business users.
- Developed complex SQL queries in Athena for ad-hoc analysis and reporting.
- Integrated Tableau with Athena for real-time data visualization and reporting.
- Worked on data performance tuning in Athena, optimizing query execution plans.
- Created Tableau dashboards and reports to provide actionable insights to business stakeholders.
- Collaborated with cross-functional teams to identify business requirements and translate them into data engineering solutions.

Environment: Agile, AWS glue, S3, Amazon Athena, lambda functions, Pyspark, PL/SQL, XML, JSON, Avro, CSV, Parquet, Tableau, MySQL

Client: Wells Fargo, India
Oct 2020 – Nov 2022
Role: Azure Data Engineer

Responsibilities:

- Analyze, design, and build Modern data solutions using Azure PaaS service to support visualization of data. Understand the current Production state of an application and determine the impact of new implementation on existing business processes.
- Architect & implement medium to large-scale BI solutions on Azure using Azure Data Platform services (Azure Data Lake, Data Factory, Data Lake Analytics, Stream Analytics, Azure SQL DW, HDInsight/Databricks, NoSQL DB, Cosmos DB).
- Established a formal EDM, MDM (Master Data Management) program that creates effective engagement between Business operation, EDM, delivery team, and IT.
- Designed and developed SSIS (ETL) packages to validate, extract, transform, and load data from the OLTP system to the Data warehouse.
- Designed and implemented Tables, Functions, Stored Procedures, and Triggers in SQL Server 2016 and wrote the SQL.
- Demonstrated proficiency in Agile methodologies, working within cross-functional Agile teams to deliver data engineering solutions on schedule and within scope.
- Designed and implemented end-to-end data solutions on the Azure cloud platform, including Azure Databricks, Azure Synapse Pipeline, and Azure Blob Storage.
- Developed and managed Azure Data Lake and Azure Blob Storage accounts to store, manage, and secure data assets.
- Worked extensively with Azure SQL Database, incorporating PL/SQL for data modeling, stored procedures, and triggers.
- Implemented database triggers and stored procedures to automate data processes and ensure data integrity.
- Created and maintained ETL pipelines using Azure Data Factory to orchestrate data workflows.
- Collaborated with cross-functional teams to understand business requirements and translated them into scalable data solutions using Azure services.
- Designed and developed complex data models and database schemas in Azure SQL Database, optimizing data storage, retrieval, and organization.
- Engineered end-to-end data pipelines using SQL for data extraction, transformation, and loading (ETL) processes, ensuring high-quality data for analytics and reporting.
- Conducted data cleansing, transformation, and enrichment using SQL scripts to ensure data accuracy and consistency.
- Implemented data integration processes, leveraging PL/SQL, to move and transform data between on-premises and Azure environments.
- Utilized Azure Data Factory or similar tools for orchestrating data workflows, integrating PL/SQL scripts seamlessly.
- Utilized Azure Monitor and other performance monitoring tools to identify and address bottlenecks in data processing pipelines.
- Created and optimized SQL queries, stored procedures, and data processing logic for data analysis and reporting within Azure SQL Database.
- Utilized Spark Streaming and Azure Databricks for real-time data processing and streaming analytics.
- Developed end-to-end data pipelines in Azure Databricks, encompassing the bronze, silver, and gold stages for comprehensive data processing.
- Implemented Spark Streaming for real-time data processing and analytics, enabling immediate insights from streaming data sources.
- Created and maintained historical data models and schemas in Snowflake, ensuring data organization and indexing for efficient querying.
- Leveraged Terraform to automate infrastructure provisioning, enabling consistent and repeatable deployments in Azure.
- Implemented CI/CD pipelines using GitHub DevOps to automate testing and deployment of data solutions.
- Collaborated with Azure DevOps teams to ensure high availability, scalability, and resource optimization for data systems.
- Successfully completed data migration projects to Azure, ensuring data consistency and integrity during the transition.

- Automated the deployment of data engineering code and pipelines using GitHub Actions to enhance efficiency and consistency.
- Strong skills in visualization tools Power BI, Confidential Excel - formulas, Pivot Tables, Charts, and DAX Commands.
- Worked on creating a few Power BI dashboard reports and heat map charts and supported numerous dashboards, pie charts, and heat map charts.

Environment: Python, SQL, PL/SQL, Azure Databricks, Azure synapse pipeline, Azure blob storage, Azure data lake, Azure SQL, Azure SQL, Spark streaming, GitHub (DevOps), Terraforms, Snowflake, PySpark, Pandas

Client: LTIMindtree
Role: Data Analyst
Aug 2018 – Sep 2020

Responsibilities:
- Designed & Created Test Cases based on the Business requirements (Also referred to as Source to Target Detailed mapping document & Transformation rules document).
- Involved in extensive DATA validation using SQL queries and back-end testing.
- Designed DTS/SSIS packages to transfer data between servers, load data into the database, archive data files from different DBMSs using SQL Enterprise Manager/SSMS on SQL Server 2008 environment, and deploy the data.
- Created SSIS packages to capture the daily maintenance plan scheduled jobs status success failure with a status report daily.
- Worked with SQL Server and T-SQL in constructing DDL/DML triggers, tables, user-defined functions, views, indexes, Stored Procedures, user profiles, relational database models, cursors, Common Table Expression CTEs, data dictionaries, and data integrity.
- Designed ETL Extract, Transform, and Load strategy to transfer data from source to stage and stage to target tables in the data warehouse tables and OLAP database from heterogeneous databases using SSIS and DTS Data Transformation Service.
- Developed SSIS packages to export data from Excel
- Spreadsheets to SQL Server, automated all the SSIS packages, and monitored errors using SQL Job daily.
- Prepared reports using SSRS SQL Server Reporting Service to highlight discrepancies between customer expectations and customer service efforts which involved scheduling the subscription reports via the subscription report wizard.
- Involved in generating and deploying various reports using global variables, expressions, and functions using SSRS.
- Migrated data from legacy system text-based files, Excel spreadsheets, and Access to SQL Server databases using DTS, SQL Server Integration Services SSIS to overcome the transformation constraints.
- Automated the SSIS jobs in the SQL scheduler as SQL server agent jobs for daily, weekly, and monthly loads.

Environment: T-SQL, SQL Server 2008/ 2008R2, SSRS, SSIS, SSAS, MS Visio, BIDS, Agile.